

Emotion Recognition From Text using BiLSTM + Self-Attention + GloVe (NO CNN)

Deep Learning



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INTRODUCTION

Emotion recognition from text is a fundamental task in Natural Language Processing (NLP) with applications in sentiment analysis, social media monitoring, and opinion mining. One particularly challenging aspect of emotion recognition is identifying non-literal expressions such as **sarcasm and irony**, where the intended meaning often contradicts the literal wording of the sentence.

Twitter text poses additional challenges due to its short length, informal language, and lack of explicit context. Traditional machine learning approaches often struggle to capture such implicit meanings. Therefore, this project applies a **deep learning approach based on Bidirectional Long Short-Term Memory (BiLSTM) and self-attention mechanisms** to recognize emotion-related categories from tweet text.

The objective of this project is to classify tweets into three emotion-related classes: **regular, irony, and sarcasm**, using contextual sequence modeling and semantic word representations.

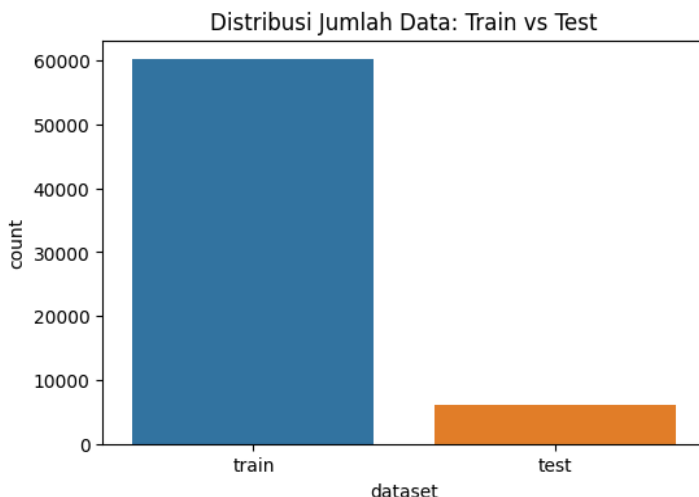
1. Dataset

The dataset consists of Twitter text data stored in two files: train.csv and test.csv. Each tweet in the training set is associated with an emotion-related label.

1.1. Dataset Attributes

Tweets = Raw tweet text

Class = Label Category (*sarcasm, regular, irony*)



1.2. Class Distribution

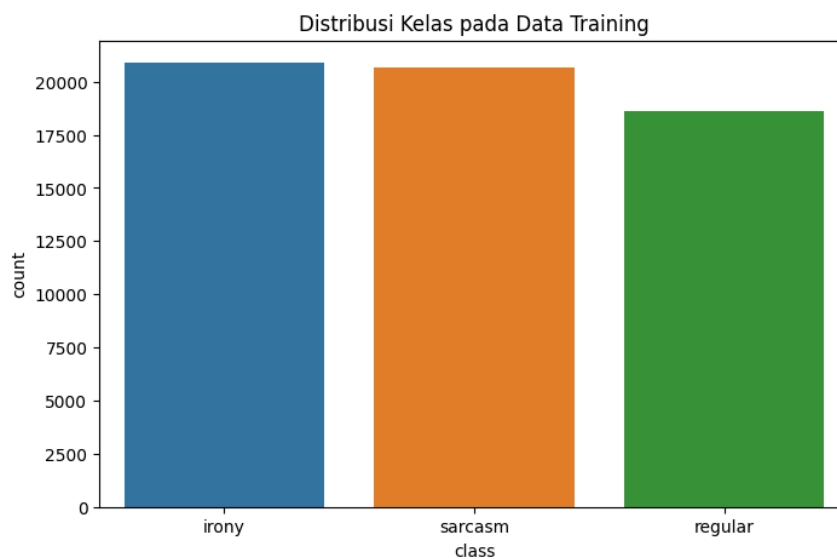
The original dataset contained four classes:

- Regular
- irony
- sarcasm
- figurative

To reduce semantic ambiguity and improve classification clarity, the **figurative class was removed** from both training and testing datasets. The final model focuses on three classes:

- **regular**
- **irony**
- **sarcasm**

Class imbalance is present in the dataset; therefore, class weighting is applied during training.



1.3. Preprocessing

The following preprocessing steps were applied to clean and normalize the text data:

- Conversion of text to lowercase
- Removal of URLs and website links
- Removal of user mentions (e.g., @username)
- Removal of hashtags to avoid label leakage
- Removal of extra whitespace

The cleaned text was tokenized using Keras Tokenizer. All sequences were padded or truncated to a fixed length of 80 tokens, ensuring consistent input size for the neural network.

2. Modelling

2.1 Word Representation

This project uses **pre-trained GloVe word embeddings (glove.6B.200d)** to represent words as dense vectors. The embedding layer:

- Has a dimension of 200
- Is initialized using GloVe vector.
- Is set as **non-trainable** to preserve semantic information

Words not found in the GloVe vocabulary are initialized with zero vectors.

2.2 Model Architecture

The proposed model architecture is designed to capture both sequential context and word-level importance.

Input Text

→ Embedding Layer (GloVe)

→ SpatialDropout1D

→ Bidirectional LSTM

→ Self-Attention

→ Global Max Pooling

→ Dense (ReLU)

→ Dropout

→ Softmax Output

Explanation:

BiLSTM captures contextual dependencies in both forward and backward directions

Self-attention allows the model to focus on emotionally relevant words

Global max pooling reduces sequence output into a fixed-size representation

Dense layers perform final classification

2.2 Training Configuration

Parameter	Value
Optimizer	Adam
Learning Rate	3e-4
Batch Size	64
Epochs	30
Loss Function	Sparse Categorical Crossentropy
Max Sequence Length	80

To improve robustness, **EarlyStopping**, **ReduceLROnPlateau**, and **ModelCheckpoint** callbacks were applied during training.

3. Evaluation

3.1 Metrics

The model was evaluated using multiple performance metrics:

- **Accuracy**
- **Macro F1-score**
- **Weighted F1-score**
- **Confusion Matrix**

3.2 Results Summary

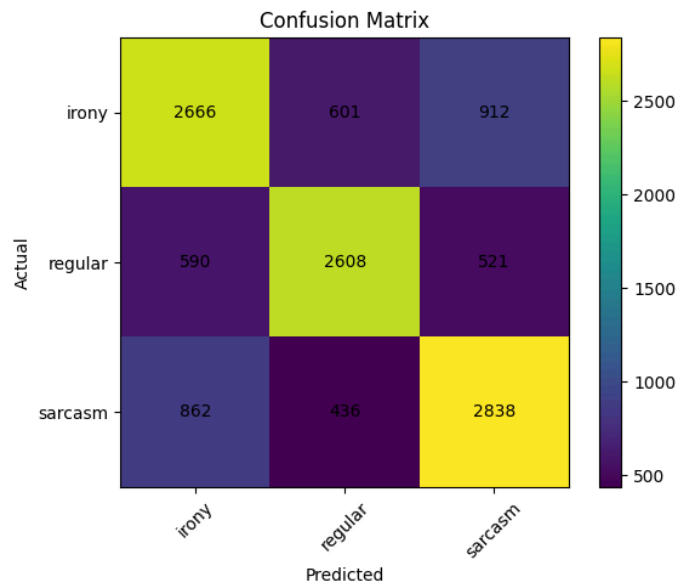
The trained model achieved:

- **Accuracy:** 67%
- **Macro F1-score:** ~0.675
- **Weighted F1-score:** ~0.674

The *regular* class achieved the highest performance, while most misclassifications occurred between *irony* and *sarcasm*, which is expected due to their semantic similarity.

3.3 Confusion Matrix

Classification report:				
	precision	recall	f1-score	support
irony	0.65	0.64	0.64	4179
regular	0.72	0.70	0.71	3719
sarcasm	0.66	0.69	0.68	4136
accuracy			0.67	12034
macro avg	0.68	0.68	0.68	12034
weighted avg	0.67	0.67	0.67	12034



The performance of the proposed BiLSTM with Attention model was evaluated using precision, recall, F1-score, and a confusion matrix on the validation dataset consisting of 12,034 samples. The model achieved an overall accuracy of **67%**, with a **macro-averaged F1-score of 0.68** and a **weighted F1-score of 0.67**, indicating balanced performance across classes despite moderate class overlap.

Among the three emotion classes, **regular** achieved the best performance with an F1-score of **0.71**, suggesting that neutral or non-figurative expressions are more easily distinguishable by the model. The **sarcasm** class obtained an F1-score of **0.68**, while **irony** showed slightly lower performance with an F1-score of **0.64**, reflecting the inherent linguistic ambiguity of ironic expressions.

The confusion matrix reveals that the most frequent misclassifications occur between **irony and sarcasm**, which is expected due to their semantic similarity and shared contextual cues. Specifically, 912 irony samples were misclassified as sarcasm, and 862 sarcasm samples were predicted as irony. This overlap highlights the challenge of distinguishing subtle pragmatic differences using textual cues alone.

Overall, the results demonstrate that the attention-enhanced BiLSTM model is effective for emotion recognition from text, while also indicating room for improvement in handling nuanced figurative language. Future work may benefit from contextualized embeddings or transformer-based architectures to further improve discrimination between closely related emotional categories.

4. Deployment

4.1 Manual Prediction Console

Notebook includes CLI mode:

```
=== MODE PREDIKSI MANUAL ===
```

Masukkan kalimat: ...

Prediksi kelas: sarcasm

Probabilitas:

sarcasm : 0.90

irony : 0.08

regular: 0.02

4.2 Prediction Function

```
def predict_emotion(text: str):
```

```
    """
```


Input: text mentah (string)

Output: (label_pred, probs_array)

```
"""
```

```
cleaned = clean_text(text)
```

```
seq = tokenizer.texts_to_sequences([cleaned])
```

```
pad = pad_sequences(seq, maxlen=max_len, padding='post', truncating='post')
```

```
probs = model.predict(pad)[0]
```

```
predid = probs.argmax()
```

```
pred_label = id2label[pred_id]
```

```
return pred_label, probs
```

4.3 Streamlit

URL: <https://deep-learning-project.streamlit.app/>

Model Playground (Prediction)

Architecture: BiLSTM + Attention Mechanism

Enter text:

Yeah, this meeting could not have been more exciting

Predict

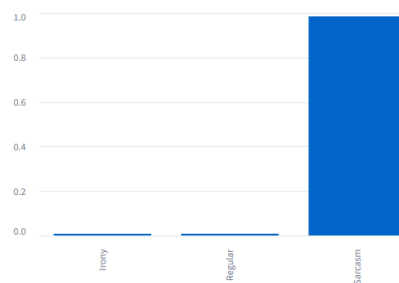
Prediction: **SARCASM**

Confidence Score

98.48%

> View Preprocessing Result

Class Probabilities



The trained model was further deployed using a Streamlit-based interface to facilitate real-time inference testing beyond the notebook environment. The deployment enables users to input arbitrary text samples and immediately observe predicted sarcasm probability scores.

The interface showcases:

- a text input field,
- predicted label output,
- full probability distribution across classes.

This deployment demonstrates not only the feasibility of sarcasm classification in controlled laboratory settings but also its practical usability in lightweight web applications aimed at content moderation, sentiment analytics, and social media monitoring.

5. Reflection

- Sarcasm detection is challenging due to implicit meaning.
- BiLSTM with attention captures context effectively.
- Removing the figurative class improved model focus.

Future work should consider:

- Fine-tuning transformer-based models such as BERT or RoBERTa
- Incorporating emoji and sentiment-aware features
- Leveraging conversational or user-level context

6. Conclusion

This study demonstrates that **BiLSTM combined with self-attention and GloVe embeddings** is an effective approach for emotion recognition from Twitter text. The model achieves balanced performance across classes and generalizes well without relying on transformer-based architectures.

The results confirm that sequence-based deep learning models remain a strong baseline for emotion recognition tasks.

7. References

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<https://www.kaggle.com/datasets/nikhiljohnk/tweets-with-sarcasm-and-irony>
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